

# Athul V Nair

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GitHub: github.com/athul-v-nair Website: Athul V Nair - Portfolio

## Professional Summary

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AI Engineer with a strong foundation in machine learning and deep learning systems, transitioning from a professional software engineering background. Passionate about building AI-powered applications using Generative AI frameworks and large language models. Actively developing hands-on projects that explore real-world use cases of GenAI systems.

## Technical Skills

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**AI Technologies:** Large Language Models (LLMs), Generative AI (GenAI), Transformers, Retrieval-Augmented Generation (RAG), Vectorless RAG, Agentic AI, Prompt Engineering, AI Evaluation, AI Observability

**AI/ML Frameworks:** LangChain, LangGraph, PyTorch, Scikit-learn

**Programming Languages:** Python, JavaScript, TypeScript, SQL, PHP

**Backend & APIs:** FastAPI, REST APIs, JSON, XML

**Data Science & Analytics:** NumPy, Pandas, Matplotlib, Seaborn

**Databases:** MySQL, MongoDB, ChromaDB

**Cloud & Tools:** LangSmith, Docker, Git, Postman, Jira, HuggingFace, AWS(Bedrock,Transcribe,S3), Adobe Experience Cloud.

**Web Technologies:** React.js, Laravel

**Methodologies:** Agile, Scrum

## Projects

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### JobAssist — AI Resume & Career Agent

GitHub Repository

- Built an end-to-end **agentic AI pipeline** using **LangGraph** that autonomously analyzes a resume against a job description, rewrites it for ATS optimization, and generates a full interview preparation kit.
- Designed a **multi-node LangGraph workflow** with conditional routing — an **LLM-as-judge rescorer** iterates the rewrite loop until ATS score ( $\geq 80$ ) and match percentage ( $\geq 75$ ) thresholds are met, with a configurable max-iteration cap.
- Generated tailored **cover letters**, **skill gap analysis**, prioritized **learning roadmaps** with specific resources, and **interview Q&A** grouped by topic — all grounded in the candidate's actual experience.
- Built an **interactive Streamlit UI** with a live pipeline progress sidebar and tabbed result views.
- **Tech Stack:** LangGraph, LangChain, LangSmith, NVIDIA NIM, PyMuPDF, Pydantic, Streamlit

### Portfolio AI Chat Assistant

GitHub Repository

- Built a **Retrieval-Augmented Generation (RAG)** chatbot for a portfolio website, enabling context-aware question answering over personal and project data.
- Designed and implemented end-to-end pipelines for **ingestion (parsing, chunking, embedding, vector storage)**, **retrieval(query embedding, chat history)** and **generation**.
- Implemented **session-based conversational memory**, allowing responses to leverage historical chat context for improved coherence.
- Developed scalable backend API using **FastAPI** and integrated seamlessly with frontend.
- Evaluated system using retrieval and generation metrics, achieving **Recall@3: 1.0, MRR: 1.0**, with **Context Precision: 0.50**
- **Tech Stack:** LangChain, ChromaDB, FastAPI, Docker, Git

### Immersify – AI Powered Immersive Reading Experience

Immersive Reading Experience for Kids & Visually Impaired

- Developed an **AI-powered application** that generates immersive sound experiences during read-aloud sessions.
- Implemented **real-time identification of sound producing words** to trigger contextual audio effects based on detected words (e.g., lion roars when "lion" is read).

- Developed an AI-powered immersive reading application that received positive feedback from judges and participants during a hackathon evaluation.
- Built the backend using **AWS Bedrock** for model integration and **AWS Transcribe** for speech-to-text processing, while storing application data in **AWS S3**.

### Contrack - Contract Management System

Enterprise contract management and tracking system.

- Built a **full-stack SaaS platform** to manage enterprise contracts and track revenue projections.
- Implemented **CRUD operations, role-based access control, and Microsoft SSO authentication**.
- Designed analytics dashboards for monitoring contract revenue and business insights.
- **Tech Stack:** React.js, Laravel, PHP, MySQL, REST APIs.

## Professional Experience

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### Software Engineer

Apr 2025 – Present

Experion Technologies — Ernakulam, Kerala

- Delivered **web analytics solutions** for a large US-based automobile manufacturer using Adobe Experience Platform, supporting marketing campaigns reaching **2M+ monthly visitors**.
- Built website-wide **event tagging infrastructure** enabling tracking of **1000+ custom user interactions**, providing granular insights into customer behavior patterns.
- Integrated multiple **third-party marketing and analytics pixels** to support campaign performance measurement.
- Built and analyzed **analytics dashboards and traffic insights** to support product and marketing decisions.
- Collaborated closely with stakeholders and clients to deliver scalable analytics implementations.
- Contributed to securing **2 additional projects** through effective solution delivery and client satisfaction.

### Associate Software Engineer

May 2024 – Apr 2025

Experion Technologies — Ernakulam, Kerala

- Worked across multiple international projects delivering production-ready web applications.
- Integrated CRM systems with product platforms to improve customer acquisition workflows.
- Maintained and optimized production systems by resolving bugs and improving application performance.
- Migrated legacy technology stacks to newer frameworks, improving maintainability and system performance.
- Received **team appreciation** for delivering a critical migration that enhanced platform stability.

### Trainee Software Engineer

Oct 2023 – May 2024

Experion Technologies — Trivandrum, Kerala

- Trained in full-stack web development and built production-ready applications.
- Developed and deployed a **Contract Management System (Contrack)** for tracking enterprise contracts and revenue projections.
- Created and released **10+ REST API endpoints** with comprehensive error handling and validation.
- Collaborated with **7-member team** to deliver the platform receiving positive feedback from stakeholders.

## Education

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### SCMS School of Engineering and Technology

Jul 2019 – Jun 2023

Bachelor of Technology, Computer Science and Engineering

GPA: 3.25

Relevant Courses: Artificial Intelligence, Machine Learning, Database Management Systems, Data Structures & Algorithms